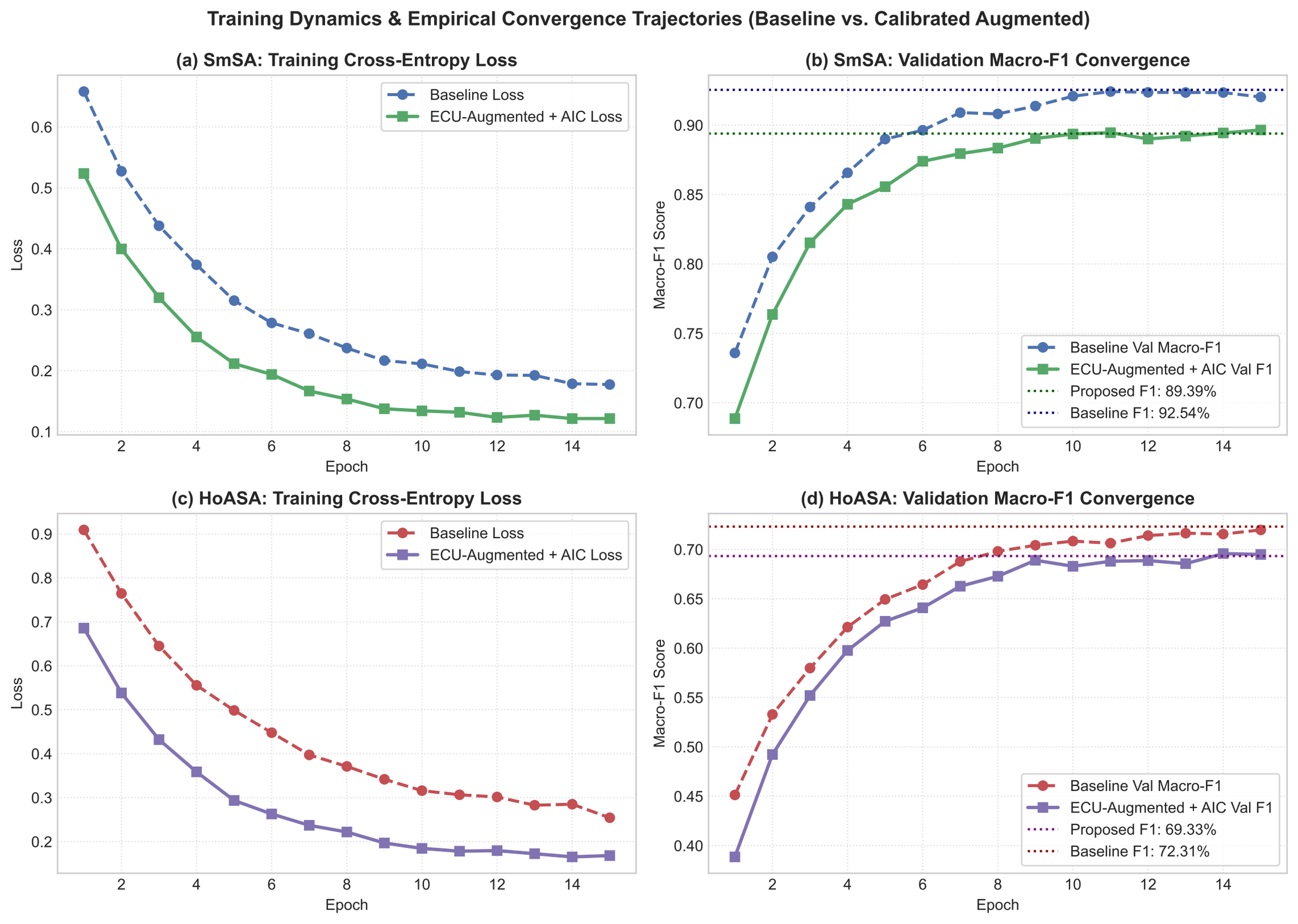
Training Dynamics & Empirical Convergence Trajectories (Baseline vs. Calibrated Augmented)
(a) SmSA: Training Cross-Entropy Loss
Baseline Loss
ECU-Augmented + AIC Loss
Loss
Epoch
(b) SmSA: Validation Macro-F1 Convergence
Macro-F1 Score
Baseline Val Macro-F1
ECU-Augmented + AIC Val F1
Proposed F1: 89.39%
Baseline F1: 92.54%
Epoch
(c) HoASA: Training Cross-Entropy Loss
Baseline Loss
ECU-Augmented + AIC Loss
Loss
Epoch
(d) HoASA: Validation Macro-F1 Convergence
Macro-F1 Score
Baseline Val Macro-F1
ECU-Augmented + AIC Val F1
Proposed F1: 69.33%
Baseline F1: 72.31%
Epoch